STUDENT PERFORMANCE FACTOR ANALYSIS

A Summer Internship Report submitted in partial fulfillment of the requirements for the award of degree of

BACHELOR OF TECHNOLOGY
In
DEPARTMENT OF CSE– AI&ML

Submitted by
Chinta Surya Harshita
22P31A4276



ADITYA COLLEGE OF ENGINEERING & TECHNOLOGY (A)

Approved by AICTE, Permanently affiliated to JNTUK & Accredited by NAAC with 'A^{+}' Grade

Recognized by UGC under the sections 2(f) and 12(B) of the UGC Act 1956

Aditya Nagar, ADB Road –Surampalem 533437, East Godavari Dist., A.P.

2025-26

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Aditya Nagar, ADB Road –Surampalem 533437, East Godavari Dist.,A.P.

CSE- AI&ML

CERTIFICATE

This is to certify that the Internship report entitled *"***STUDENT PERFORMANCE FACTOR ANALYSIS***"* is being submitted by **Chinta Surya Harshita (22P31A4276)** In partial fulfillment of the requirements for the award of the B.Tech degree in **CSE – AI&ML** for the academic year 2025-2026.

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Mr.D.Satyanarayana, **M.Tech**
Assistant Professor
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Head of the Department
Dr.Naga Bhargavi **M.Tech. , Ph.D**
Associate Professor
CSE-AI&ML

DECLARATION

I hereby declare that the Internship entitled **"STUDENT PERFORMANCE FACTOR ANALYSIS"** is a genuine report. This work has been submitted to the **ADITYA COLLEGE OF ENGINEERING & TECHNOLOGY (A),** Surampalem, permanently affiliated to **JNTUK, KAKINADA** in partial fulfillment of the **B.Tech** degree**.**

I also hereby declare that this internship report not submitted in full or partial any other university for any degree.

CHINTA SURYA HARSHITA
(22P31AA4276)

ADITYA COLLEGE OF ENGINEERING & TECHNOLOGY

An AUTONOMOUS Institution

Approved by AICTE, Permanently Affiliated to JNTUK, Accredited by NBA & NAAC with A+ Grade
Recognized by UGC under Sections 2(f) and 12(B) of UGC Act, 1956
Aditya Nagar, ADB Road, Surampalem, Kakinada District - 533 437, A.P.
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VISION

To induce higher planes of learning by imparting technical education with

- ✓ International standards
- ✓ Applied research
- ✓ Creative Ability
- ✓ Value based instruction and to emerge as a premiere institute

MISSION

Achieving academic excellence by providing globally acceptable technicaleducation by forecasting technology through

- ✓ Innovative Research And development
- ✓ Industry Institute Interaction
- ✓ Empowered Manpower

Principal
PRINCIPAL
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Engineering & Technology
SURAMPALEM- 533 437

DEPARTMENT OF CSE - AIML

VISION & MISSION OF THE DEPARTMENT

VISION

To be a recognized Computer Science and Engineering hub striving to meet the growing needs of the Industry and Society.

MISSION

M1: Imparting Quality Education through state-of-the-art infrastructure with industry collaboration.

M2: Enhance Teaching Learning Process to disseminate knowledge.

M3: Organize Skill based, Industrial and Societal Events for overall Development

PROGRAMME EDUCATIONAL OBJECTIVES (PEO's)

PEO1:	Proficiency in AI & ML. Applications Expertise in designing, developing and deploying AI and ML solutions for healthcare, finance, and autonomous system problems
PEO2:	Lifelong Learning a and Adaptability Adapt to evolving technologies and industry trends in AI and M. through continuous learning and self-improvement
PEO3:	Effective Communication and Collaboration Possess strong communication mating them to effectively collaborate with interdisciplinary teams and solve complex problems using AI and ML concepts

PROGRAMME SPECIFIC OUTCOMES (PSO's)

PSO1:	**AI and ML system development** Design and develop, and implement AI and ML systems by applying fundamental principles, algorithms and techniques
PSO2:	**Data-Driven Decision-Making** Collect, Preprocess and analyze large and diverse datasets to extract meaningful insights using AI and ML techniques to support data-driven decision-making processes in various domains.

HEAD OF THE DEPARTMENT

PROGRAMME OUTCOMES (POS)

After successful completion of the program, the graduates will be able to

PO1:	**Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2:	**Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO3:	**Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO4:	**Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO5:	**Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO6:	**The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PO7:	**Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
PO8:	**Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO9:	**Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO10:	**Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11:	**Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12:	**Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.



ANDHRA PRADESH STATE SKILL DEVELOPMENT CORPORATION

DEPARTMENT OF SKILLS DEVELOPMENT & TRAINING

GOVERNMENT OF ANDHRA PRADESH

CERTIFICATE

OF COMPLETION

CERTIFICATE NO: APSSDC/25/INT/DA-1013

This is to certify that CHINTA SURYA HARSHITA,

bearing Roll Number 22P31A4276, *from*

ADITYA COLLEGE OF ENGINEERING AND TECHNOLOGY, *has*

successfully completed the course DATA ANALYSIS USING PYTHON

under the ***Summer Online Internship Program - 2025*** *conducted by*

the ***Andhra Pradesh State Skill Development Corporation (APSSDC)***

from 12TH MAY 2025 *to* 12TH JULY 2025.

ED - TECHNICAL
(K. RAGHU)

MD & CEO
(G. GANESH KUMAR, I.A.S.)

ACKNOWLEDGEMENT

I offer my sincere thanks to dynamic and beloved Principal, **Dr. A. Ramesh**, **M.Tech., Ph.D , Aditya College of Engineering & Technology(A)** for his co-operation.

My sincere thanks to **Dr.Naga Bhargavi M.Tech., Ph.D ,** Head of the department of **CSE-AI&ML** for her valuable support.

I express my sincere gratitude to my internship guide, **Mr.D.Satyanarayana, M.Tech,** Department of **CSE-AI&ML**, for his valuable guidance and encouragement which has been helpful in successful completion of this internship.

With immense pleasure I would like to express my deep sense and heart full thanks to the management of Aditya College of Engineering & Technology (A).

Learning Objectives/Internship Objectives

1.Data Handling and Preprocessing:

- Import data from sources like Google Drive using Python.
- Clean the data by removing duplicates and fixing missing or incorrect values.
- Use datetime tools to manage time-based data like attendance or study hours.

2.Exploratory Data Analysis (EDA):

- Used charts and graphs to see patterns in student data.
- Checked which factors (like sleep, attendance) are linked to better exam scores.
- Looked at data spread to find outliers and unusual values.
- Compared groups (like boys vs. girls or public vs. private school) to see differences in performance.

3. Data Visualization Skills:

- Created graphs and charts (like bar plots, pie charts, and scatter plots) to make data easy to understand.
- Used heatmaps to show how different factors relate to student performance.
- Plotted trends over time, such as changes in attendance or scores.
- Highlighted key insights using colors and labels to improve clarity for decision-makers.

4.Time-Series Analysis:

- Tracked student data over time, such as exam scores or attendance across months or terms.
- Used date and time features with Python's datetime to organize and analyze time-based data.
- Plotted trends to see how performance changes during the school year.

5. Insight Generation for Real-World Applications:

- Identified key factors (like motivation and parental support) that strongly impact student performance.
- Provided actionable insights for teachers, parents, and policymakers to support student success.
- Suggested data-driven interventions, such as improving sleep habits or boosting attendance.
- Helped personalize learning by understanding each student's unique challenges and strengths.

6.Practical Implementation of Data Science Concepts:

- Applied data cleaning, preprocessing, and feature engineering on real student data.
- Used statistical tools and machine learning to analyze and predict academic performance.
- Built visualizations to clearly communicate patterns and findings.
- Turned raw data into meaningful insights that can solve real-world education problems.

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CHAPTER 1: EXECUTIVE SUMMARY

This study explored various factors influencing student academic performance using data analysis and visualization techniques. By examining variables such as attendance, parental support, sleep duration, motivation, and others, the research aimed to identify key predictors of academic success. Using Python libraries like pandas, matplotlib, and seaborn, the analysis uncovered important relationships and patterns in student data.

Learning Objectives and Outcomes:

1. **Objective:** Gain practical experience with Python tools for data analysis.
   Outcome: Effectively applied pandas, matplotlib, and seaborn for data cleaning, transformation, and visualization.
2. **Objective:** Understand and implement data preprocessing techniques.
   Outcome: Handled missing values, removed duplicates, encoded categorical variables, and managed time-based data.
3. **Objective:** Conduct exploratory data analysis (EDA) to discover trends.
   Outcome: Used correlation analysis and visualizations such as heatmaps and scatter plots to examine relationships between factors and academic scores.
4. **Objective:** Identify key factors influencing academic achievement.
   Outcome: Found that attendance, parental involvement, motivation, and sleep had significant positive effects on performance.
5. **Objective:** Develop actionable insights for educators and policymakers.
   Outcome: Provided data-driven recommendations to support student success through targeted interventions.

Sector of Business and Organization:

This study contributes to the education and data analytics sector, focusing on improving learning outcomes through data-driven insights. The research highlights how integrating behavioral, social, and academic data can help educators tailor strategies to student needs.

Summary of Activities:

During the internship, the following activities were undertaken:

- Collected and imported student-related datasets for analysis.
- Cleaned and preprocessed data to ensure accuracy and consistency.
- Conducted exploratory data analysis on variables such as attendance, sleep, motivation, and exam scores.
- Visualized data trends and correlations using charts, heatmaps, and scatter plots.
- Applied statistical and machine learning techniques to rank the importance of factors.
- Generated practical insights to guide educational decision-making and personalized learning plans.

In summary, this research enhanced analytical and technical skills while providing meaningful insights into factors affecting student academic performance. The study demonstrates the value of data-driven approaches in education to support student achievement and well-being.

CHAPTER 2: OVERVIEW OF THE ORGANIZATION

A. Introduction of the Organization:

The study focuses on educational institutions, including public and private schools, which are key organizations in delivering academic services to students. These institutions provide structured learning environments, resources, and support systems aimed at enhancing student knowledge and skills. The data for this research was collected from multiple schools and households, reflecting diverse socioeconomic backgrounds and learning contexts.

B. Vision, Mission, and Values of the Organization:

- **Vision**: To foster an inclusive and supportive educational environment that empowers every student to reach their full academic potential.
- **Mission**: To utilize data-driven insights to improve teaching effectiveness, student engagement, and learning outcomes.
- **Core Values**: Equity in education, holistic student development, continuous improvement, collaboration among educators, and promoting well-being alongside academic success.

C. Policy of the Organization, in Relation to the Intern Role:

- Ethical handling of student data and privacy protection.
- Inclusive learning opportunities for all students regardless of background.
- Encouragement of parental involvement and community engagement.
- Promoting balanced development including academics, health, and social skills.

D. Organizational Structure:

Schools typically have a hierarchical structure designed to support teaching and learning:

- Administration: Includes principals and school leaders responsible for overall management.
- Teachers: Deliver classroom instruction and assess student progress.
- Support Staff: Counselors, tutors, and extracurricular coordinators assist students' academic and emotional needs.
- Parents and Community: Play a critical role in supporting student learning outside school.

This structure fosters collaboration among educators, families, and students to create a nurturing learning environment.

E. Roles and Responsibilities of the Employees in Which the Intern is Placed:

- The study's focus involves educators, school administrators, and support staff with responsibilities such as:
- Collecting and maintaining accurate student records.
- Monitoring attendance and academic progress.
- Providing tutoring and motivational support.
- Engaging with families to promote student well-being and involvement.
- Applying data insights to tailor teaching strategies and interventions.

F. Performance of the Organization in Terms of Turnover, Profits, Market Reach, and Market Value:

- Schools in the study showed varying levels of student achievement influenced by attendance, family support, and resources.
- Metrics such as exam scores, graduation rates, and participation in extracurricular activities were used to evaluate performance.
- Schools are continuously striving to improve learning outcomes while addressing challenges
- related to socioeconomic disparities**.**

G. Future Plans of the Organization:

- Educational institutions aim to enhance student success through:
- Integrating mental health and emotional well-being metrics into student support systems.
- Expanding access to technology and internet for remote learning.
- Implementing personalized learning plans based on data insights.
- Promoting greater family and community engagement in education.
- Using longitudinal data to track and improve academic trajectories over time.

CHAPTER 3: INTERNSHIP PART

3.1 Description of Activities/Responsibilities:

During the APSSDC Internship on Data Analysis using Python, I was engaged in a series of tasks aimed at understanding the practical applications of Python programming in handling real-world datasets. The internship was fully structured, and every week had a defined set of activities that gradually developed my technical and analytical skills.

3.2 Working Conditions:

- The internship was conducted in an online mode, making use of platforms such as Google Colab/Jupyter Notebook for executing Python code.
- Communication and coordination with mentors and peers were carried out through online channels (Google Meet, WhatsApp, and email).
- Resources like datasets, Python libraries, and documentation were provided and explained in detail by the trainers.
- The environment was flexible, but deadlines were fixed for task submissions, ensuring discipline and professional responsibility.

3.3 Weekly Work Schedule :

- **Week 1:** Introduction to Python basics, working with variables, data types, and control structures.
- **Week 2:** Introduction to data handling with Pandas, working with CSV files, data cleaning (removing duplicates, handling missing values).
- **Week 3:** Introduction to Matplotlib and Seaborn for visualizations (bar charts, histograms, scatter plots).
- **Week 4:** Advanced EDA – correlation analysis, grouping, and statistical insights**.**
- **Week 5:** Time-based analysis – converting timestamps, grouping by dates, plotting trends.

- **Week 6:** Project work on Spotify Dataset – analyzing features such as energy, danceability, and valence.
- **Week 7:** Building dashboards and detailed reports with visualization.
- **Week 8**: Final project submission and presentation of findings

3.4 Equipment and Tools Used:

- Laptop/PC with internet connection.
- Python (Anaconda / Google Colab) environment.
- **Libraries:**
    - Pandas for data manipulation.
    - Matplotlib & Seaborn for data visualization.
    - Datetime for time-based analysis.
- **Datasets:** CSV files (e.g., Spotify dataset).

3.5 Tasks Performed:

- Imported and cleaned raw datasets from Google Drive.
- Identified and removed duplicate values.
- Created scatter plots to analyze energy vs danceability vs valence.
- Plotted histograms to study the distribution of valence (mood levels).
- Developed correlation heatmaps to understand relationships between audio features.
- Performed time-series analysis on listening history to visualize mood changes over time.
- Generated insights into Top 10 Most Played Songs using bar plots.
- Prepared reports summarizing observations and findings.

WEEKLY OVERVIEW OF INTERNSHIP ACTIVITIES

ACTIVITY LOG FOR THE FIRST WEEK

	Date	Day	Brief Description of Daily Activity	Learning Outcomes
1st WEEK	12-05-2025	Monday	Online Zoom orientation, introduction to internship schedule, overview of APSSDC internship structure.	Understood program flow & expectations
	13-05-2025	Tuesday	Introduction to Python programming – IDE setup, syntax, variables.	Learned Python environment setup and simple programs.
	14-05-2025	Wednesday	Data types, operators, and expressions in Python.	Ability to use variables and perform calculations.
	15-05-2025	Thursday	Control structures – if-else, loops (for, while).	Gained problem-solving using conditions & iterations.
	16-05-2025	Friday	Functions and modular programming.	Learned to write reusable and structured code.
	17-05-2025	Saturday	Weekly Test (Python Basics) conducted online.	Tested understanding of fundamentals.

WEEKLY REPORT

WEEK – 1 (From Dt:**12-05-2025** to Dt: **17-05-2025**)

Objective of the Activity Done:

To get familiar with Python programming language basics and understand how Python is applied in data analysis.

Detailed Report:

During the first week, the training started with an orientation session introducing the internship structure and objectives. I was introduced to Python programming basics, including installation, syntax, variables, operators, and data types. Later sessions covered control flow concepts such as loops and conditional statements. The week concluded with hands-on practice on functions and Python collections (lists, tuples, and dictionaries).

This gave me a solid foundation in Python, which will help in applying it to data analysis tasks in upcoming weeks.

ACTIVITY LOG FOR THE SECOND WEEK

	Date	Day	Brief Description of Daily Activity	Learning Outcomes
2nd WEEK	19-05-2025	Monday	Lists, tuples, sets, dictionaries – introduction & operations.	Learned to store & manipulate collections
	20-05-2025	Tuesday	String operations, slicing, methods, formatting.	Improved handling of text-based data.
	21-05-2025	Wednesday	File handling in Python (text & CSV)	Learned to read/write external files.
	22-05-2025	Thursday	Exception handling & debugging.	Understood how to handle runtime errors.
	23-05-2025	Friday	Practice exercises, mini problems on Python basics	Strengthened programming confidence.
	24-05-2025	Saturday	Weekly Test (Advanced Python).	Evaluated progress and understanding.

WEEKLY REPORT

WEEK – 2 (From Dt: **19-05-2025** to Dt: **24-05-2025**)

Objective of the Activity Done:

To strengthen Python fundamentals and gain problem-solving experience.

Detailed Report:

In the second week, we continued with Python, focusing more on functions, string operations, and error handling. Simple mini-tasks and exercises were given to apply the concepts learned. By the end of the week, I felt confident in writing small Python programs and applying logic.

ACTIVITY LOG FOR THE THIRD WEEK

	Date	Day	Brief Description of Daily Activity	Learning Outcomes
3rd WEEK	26-05-2025	Monday	Monday Introduction to NumPy, arrays, and differences from lists.	Learned importance of NumPy in data handling.
	27-05-2025	Tuesday	Array indexing, slicing, and reshaping.	Improved skills in accessing/modifying arrays.
	28-05-2025	Wednesday	Mathematical & statistical functions in NumPy.	Applied vectorized operations.
	29-05-2025	Thursday	Broadcasting, aggregate functions.	Understood advanced array computations.
	30-05-2025	Friday	Hands-on problems using NumPy arrays.	Strengthened numerical computing skills.
	31-05-2025	Saturday	Weekly Test (NumPy Concepts)	Tested efficiency in using NumPy.

WEEKLY REPORT

WEEK – 3 (From Dt: **26-05-2025** to Dt: **31-05-2025**)

Objective of the Activity Done:

To learn data manipulation and analysis using the Pandas library.

Detailed Report:

This week introduced Pandas, a powerful data analysis library in Python. I learned how to create Data Frames, perform indexing, filtering, grouping, and data cleaning. Practical exercises included importing CSV files, checking dataset info, handling missing values, and removing duplicates. This formed the basis for working with real-world datasets.

ACTIVITY LOG FOR THE FORTH WEEK

	Date	Day	Brief Description of Daily Activity	Learning Outcomes
4thWEEK	02-06-2025	Monday	Introduction to Pandas, Series & Data Frames.	Learned structured data representation.
	03-06-2025	Tuesday	Data cleaning, handling missing & duplicate values.	Improved preprocessing skills.
	04-06-2025	Wednesday	Indexing, selection, filtering.	Learned dataset exploration methods.
	05-06-2025	Thursday	GroupBy, aggregation, merging & joining	Understood advanced data manipulation.
	06-06-2025	Friday	Practice with CSV datasets in Pandas.	Gained confidence in handling tabular data.
	07-06-2025	Saturday	Weekly Test (Pandas Basics).	Checked learning outcomes on Pandas.

WEEKLY REPORT

WEEK – 4 (From Dt : **02-06-2025** to Dt: **07-06-2025**)

Objective of the Activity Done:

To visualize datasets using Matplotlib and Seaborn.

Detailed Report:

The focus of this week was data visualization. I learned how to plot line charts, bar charts, scatter plots, and histograms using Matplotlib. With Seaborn, I created advanced visualizations like scatter plots with hues, distribution plots, and heatmaps. These visualizations helped me understand dataset trends and correlations.

ACTIVITY LOG FOR THE FIFTH WEEK

	Date	Day	Brief Description of Daily Activity	Learning Outcomes
5thWEEK	09-06-2025	Monday	Introduction to Matplotlib – line, bar, scatter plots.	Learned to create basic charts.
	10-06-2025	Tuesday	Advanced plots – histograms, pie, stacked plots.	Understood variety of visualizations.
	11-06-2025	Wednesday	Introduction to Seaborn– aesthetics & themes.	Learned better visualization styling.
	12-06-2025	Thursday	Heatmaps, pair plots, categorical plots.	Gained advanced plotting skills.
	13-06-2025	Friday	Practice on real-world dataset visualization.	Strengthened data storytelling.
	14-06-2025	Saturday	Weekly Test (Visualization).	Evaluated plotting & analysis skills.

WEEKLY REPORT

WEEK – 5 (From Dt: **09-06-2025** to Dt: **14-06-2025**)

Objective of the Activity Done:

To explore NumPy and its use in numerical computations**.**

Detailed Report:

This week was dedicated to NumPy, where I learned about arrays, array operations, mathematical functions, and indexing. I practiced different numerical computations and matrix manipulations, which are useful for handling large datasets efficiently.

ACTIVITY LOG FOR THE SIXTH WEEK

	Date	Day	Brief Description of Daily Activity	Learning Outcomes
6thWEEK	16-06-2025	Monday	Introduction to project dataset (CSV).	Understood dataset structure.
	17-06-2025	Tuesday	Data cleaning, preprocessing.	Learned to prepare dataset for analysis.
	18-06-2025	Wednesday	Exploratory Data Analysis (EDA) with Pandas.	Discovered dataset patterns.
	19-06-2025	Thursday	Visualization of trends and distributions.	Improved data interpretation.
	20-06-2025	Friday	Documentation of initial findings.	Gained report writing skills.
	21-06-2025	Saturday	Weekly Test (EDA Concepts).	Reviewed progress in analytics.

WEEKLY REPORT

WEEK – 6 (From Dt: **16-06-2025** to Dt: **21-06-2025**)

Objective of the Activity Done:

To apply Python, Pandas, Matplotlib, and NumPy in a mini-project.

Detailed Report:

In these two weeks, I worked on the Spotify Mood Timeline Project using a dataset containing audio features like danceability , energy, valency, tempo etc.,Tasks included:

- Cleaning the dataset (removing duplicates).
- Performing exploratory data analysis (EDA).
- Creating visualizations: scatter plot, histogram, correlation heatmap, and mood trend over time.
- Identifying the Top 10 most played songs.

This project strengthened my confidence in applying all the libraries together for meaningful data insights.

ACTIVITY LOG FOR THE SEVENTH WEEK

	Date	Day	Brief Description of Daily Activity	Learning Outcomes
7thWEEK	23-06-2025	Monday	Loading Spotify dataset from Google Drive.	Learned to import & manage real data.
	24-06-2025	Tuesday	Cleaning data, removing duplicates.	Improved dataset preprocessing.
	25-06-2025	Wednesday	Scatter plot (danceability vs energy vs valence).	Understood feature relationships.
	26-06-2025	Thursday	Correlation heatmap of audio features.	Learned feature correlation analysis
	27-06-2025	Friday	Time-series mood trend & Top 10 songs visualization.	Gained storytelling with data.
	28-06-2025	Saturday	Weekly Test (Spotify Project Analysis).	Evaluated applied project knowledge .

WEEKLY REPORT

WEEK – 7 (From Dt: **23-06-2025** to Dt: **28-06-2025**)

Objective of the Activity Done:

To apply Python, Pandas, Matplotlib, and NumPy in a mini-project.

Detailed Report:

In these two weeks, I worked on the Spotify Mood Timeline Project using a dataset containing audio features like *danceability, energy, valence, tempo, etc.* Tasks included:

- Cleaning the dataset (removing duplicates).
- Performing exploratory data analysis (EDA).
- Creating visualizations: scatter plot, histogram, correlation heatmap, and mood trend over time.
- Identifying the Top 10 most played songs.

This project strengthened my confidence in applying all the libraries together for meaningful data insights.

ACTIVITY LOG FOR THE EIGHTH WEEK

	Date	Day	Brief Description of Daily Activity	Learning Outcomes
8thWEEK	30-06-2025	Monday	Final review of analysis & code.	Improved project quality.
	01-07-2025	Tuesday	Preparation of report & documentation.	Learned report writing.
	02-07-2025	Wednesday	Project presentation preparation (slides).	Developed presentation skills.
	03-07-2025	Thursday	Final Zoom presentation & discussion with mentors.	Gained confidence in public speaking.
	04-07-2025	Friday	Submission of final project (Spotify Mood Timeline).	Successfully completed internship.
	05-07-2025	Saturday	Final Test & Feedback Session.	Understood strengths & areas of improvement.

WEEKLY REPORT

WEEK – 8 (From Dt: **30-06-2025** to Dt: **05-07-2025**)

Objective of the Activity Done:

To prepare the internship report and presentation**.**

Detailed Report:

In the final week, I compiled all my activities, visualizations, and results into the final report. The mentors guided us on how to prepare professional documentation. I successfully submitted my internship report and completed the program.

CHAPTER 5: OUTCOMES DESCRIPTION

During my internship with the **Andhra Pradesh State Skill Development Corporation (APSSDC)** on Data Analysis Using Python, I had the opportunity to experience a professional learning environment that helped me grow both technically and personally. The internship not only provided academic knowledge but also gave insights into workplace culture and practices. The outcomes of the work environment are described below:

1.Work Environment and People Interactions:

The internship was conducted in a collaborative environment where trainers, mentors, and fellow interns interacted frequently. The mentors were approachable, provided constant guidance, and clarified doubts with patience. Group discussions and Q&A sessions helped me learn from peers and develop problem-solving skills through teamwork.

2. Facilities Available and Maintenance:

As the internship was conducted online, digital facilities such as Google Colab, Jupyter Notebook, Google Drive, and online meeting platforms were the main tools used. All resources, including datasets, documentation, and recorded sessions, were well-organized and accessible. The infrastructure was well-maintained in terms of uninterrupted access and timely updates.

3. Clarity of Job Roles and Responsibilities:

From the very beginning, the internship objectives and weekly schedule were clearly communicated. Each week had structured activities such as coding practice, visualization exercises, report preparation, and tests. This clarity helped me stay on track and achieve learning outcomes without confusion.

4. Protocols, Procedures, and Processes:

There was a disciplined approach to the internship. Submissions were required within deadlines, weekly reviews were conducted, and assessments ensured consistent learning. The process followed a step-by-step method: starting with Python basics, then Pandas, Matplotlib, and NumPy, and finally moving to project work.

5.Discipline and Time Management:

The internship required attending daily sessions, completing exercises, and submitting weekly tasks on time. This routine taught me the importance of time management, consistency, and self-discipline in professional life.

6.Harmonious Relationships and Socialization:

Despite being online, the program encouraged interaction among participants. Discussions, collaborative learning, and peer support created a positive and friendly learning atmosphere. I was able to socialize with students from different colleges, which broadened my perspective.

7. Mutual Support and Teamwork:

Team spirit was encouraged throughout the internship. Whenever difficulties arose, both mentors and peers provided support. This sense of mutual assistance not only solved technical issues but also motivated everyone to progress together.

8. Motivation and Encouragement:

The internship mentors regularly motivated students to explore beyond the tasks and practice additional exercises. The regular tests and project reviews encouraged healthy competition while improving our confidence.

9. Space, Ventilation, and Comfort:

Since the internship was conducted online, the work environment depended on personal study space. The flexibility of remote learning allowed me to create a comfortable learning atmosphere with good space, ventilation, and resources at home.

Conclusion:

Overall, the internship provided an excellent balance of technical training and professional environment exposure. The structured schedule, disciplined approach, and supportive mentors created an environment similar to a real workplace. This experience enhanced not only my technical expertise in data analysis but also my soft skills such as teamwork, communication, discipline, and time management.

Student Self Evaluation of the Short-Term Internship

Student Name: CHINTA SURYA HARSHITA

Registration No: 22P31A4276

Duration of Internship: 2 Months

From: 12th May 2025 **To:** 12th July 2025

Date of Evaluation:

Organization Name & Address:

Please rate your performance in the following areas:

Rating Scale: Letter grade of CGPA calculation to be provided

1	Oral communication	1	2	3	4	5
2	Written communication	1	2	3	4	5
3	Interaction ability with community	1	2	3	4	5
4	Positive Attitude	1	2	3	4	5
5	Self-confidence	1	2	3	4	5
6	Ability to learn	1	2	3	4	5
7	Work Plan and organization	1	2	3	4	5
8	Quality of work done	1	2	3	4	5
9	Time Management	1	2	3	4	5
10	Achievement of Desired Outcomes	1	2	3	4	5
OVERALL PERFORMANCE						

Signature of the Student

Evaluation by the Supervisor of the Intern Organization

Student Name: CHINTA SURYA HARSHITA

Registration No:22P31AA4276

Duration of Internship:

From: 12th May 2025 **To:** 12th July 2025

Date of Evaluation:

Organization Name & Address:

Name & Address of the Supervisor with Mobile Number

Please rate the student's performance in the following areas:

Please note that your evaluation shall be done independent of the Student's selfevaluation

Rating Scale: 1 is lowest and 5 is highest rank

1	Oral communication	1	2	3	4	5
2	Written communication	1	2	3	4	5
3	Interaction ability with community	1	2	3	4	5
4	Positive Attitude	1	2	3	4	5
5	Self-confidence	1	2	3	4	5
6	Ability to learn	1	2	3	4	5
7	Work Plan and organization	1	2	3	4	5
8	Quality of work done	1	2	3	4	5
9	Time Management	1	2	3	4	5
10	Achievement of Desired Outcomes	1	2	3	4	5
OVERALL PERFORMANCE						

Signature of the Supervisor

PHOTOS & VIDEO LINK

```
[5]: import pandas as pd
     import numpy as np
     import matplotlib.pyplot as plt
     import seaborn as sns
```

```
[6]: student=pd.read_csv("StudentPerformanceFactors.csv")
```

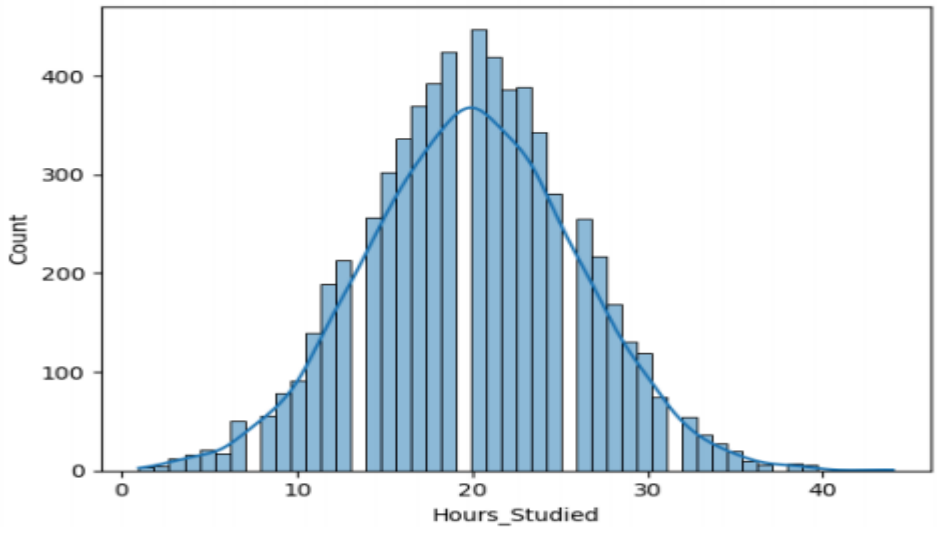
```
[7]: student
```

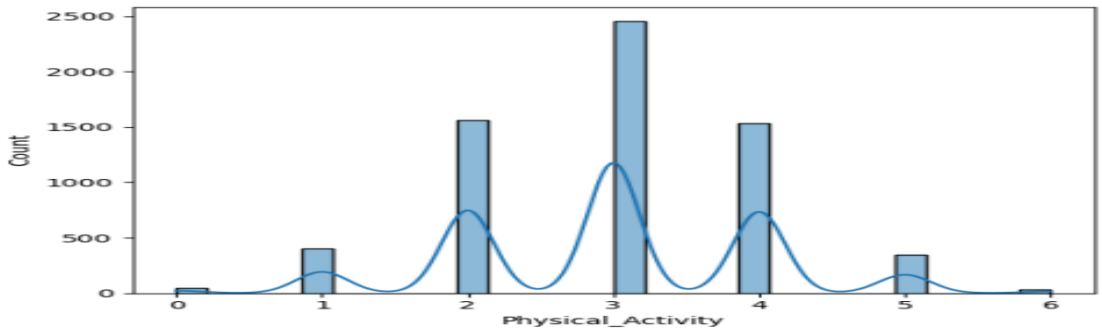
[7]:

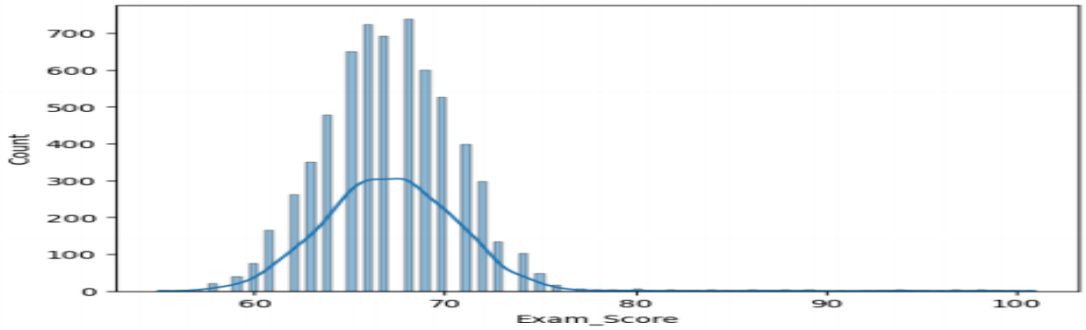
	Hours_Studied	Attendance	Parental_Involvement	Access_to_Resources	Extracurricular_Activities	Sleep_Hours	Previous_Scores	Motivation_Level	Internet_Access
0	23	84	Low	High	No	7	73	Low	Yes
1	19	64	Low	Medium	No	8	59	Low	Yes
2	24	98	Medium	Medium	Yes	7	91	Medium	Yes
3	29	89	Low	Medium	Yes	8	98	Medium	Yes
4	19	92	Medium	Medium	Yes	6	65	Medium	Yes
...	...	...	...	...	...	...	...	...	...
6602	25	69	High	Medium	No	7	76	Medium	Yes
6603	23	76	High	Medium	No	8	81	Medium	Yes
6604	20	90	Medium	Low	Yes	6	65	Low	Yes
6605	10	86	High	High	Yes	6	91	High	Yes
6606	15	67	Medium	Low	Yes	9	94	Medium	Yes

6607 rows × 20 columns

```
31]: # Loop through all columns in the DataFrame
     for col in student:
         # Check if the column is not categorical (i.e., numeric type)
         if student[col].dtype != 'O':  #  'O' stands for object
             sns.histplot(student[col],kde=True)
             #Plot histogram with Kernel Density Estimate for numeric column
             plt.show()
```

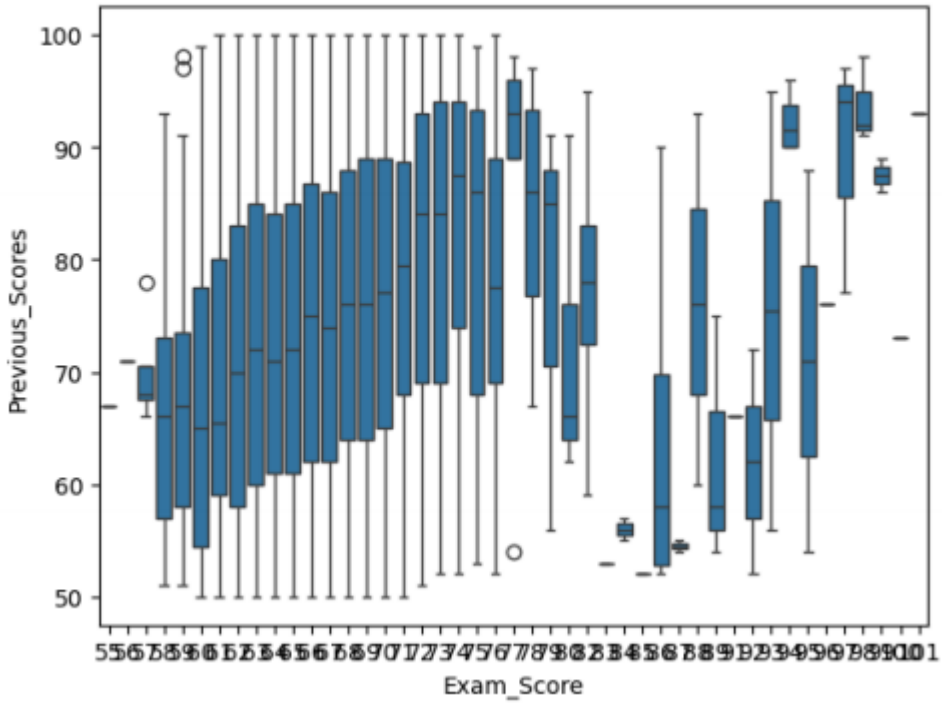


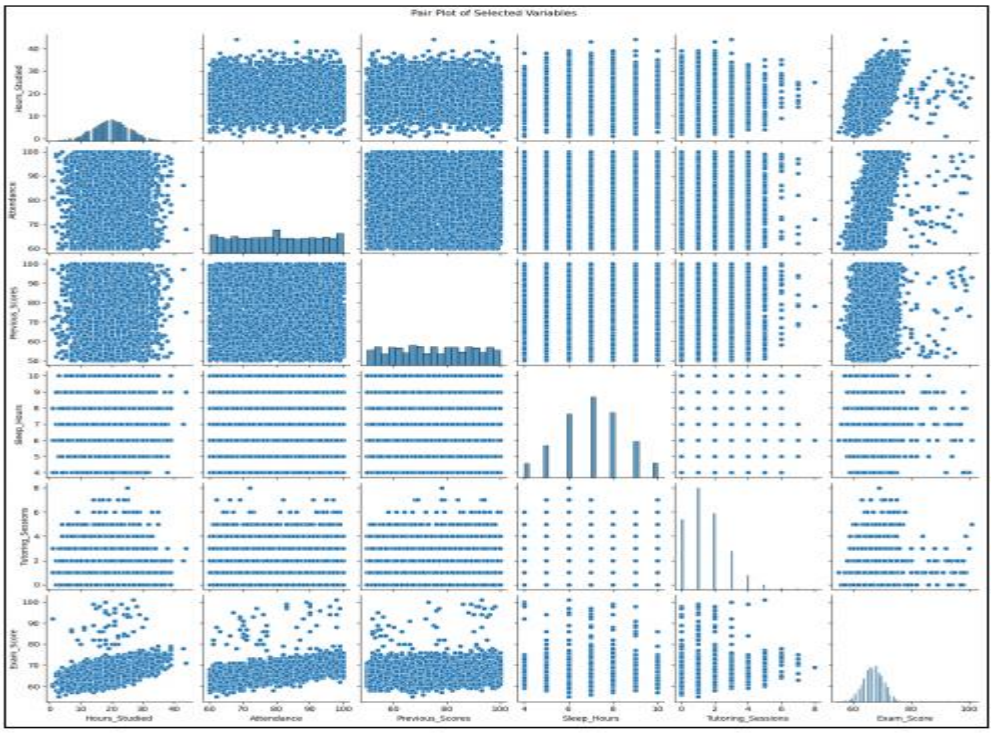





```
In [34]: sns.boxplot(x='Exam_Score',y='Previous_Scores',data=student)
```

```
Out[34]: <Axes: xlabel='Exam_Score', ylabel='Previous_Scores'>
```





VIDEO LINK:

CLICK HERE